Sai Kiran Belana

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Education

University of Connecticut *Expected Graduation: December 2025*

Masters in Computer Science Engineering GPA: 3.7/4.0

Coursework: Systems Security, Operating Systems, Computer Architecture, Cyber Security, Analysis of Algorithms, Computer Networking, Big Data Analytics

Professional Experience

Student IT Assistant, University of Connecticut *February 2024 - Present*

- Containerized university applications using Docker and deployed them on OpenShift Kubernetes clusters, achieving a 60% reduction in resource usage and enabling seamless autoscaling for dynamic workloads.
- Unified five classroom platforms into a single, scalable system with advanced filtering capabilities, reducing maintenance overhead by 80% and improving user navigation.
- Designed, developed, and deployed a OneDrive restoration tool using PHP and ReactJs; implemented CI/CD pipelines via Bitbucket and Jenkins to automate builds triggered by version control commits.

Software Engineer, LightSpeed Photonics *September 2021 - December 2023*

- Architected a RESTful web platform integrated with Slurm Workload Manager, enabling remote FPGA application deployment on HPC clusters reducing manual deployment effort by over 85%.
- Distributed tile-based matrix multiplication computation across multiple FPGAs, benchmarking OpenCL kernel performance between Xeon CPUs and FPGAs, achieving up to 10 times speedup on problem sizes 512–4096.
- Built a GUI to demonstrate data transfer between CPU & FPGA over PCIe 3.0 using FPGA FIFOs, capturing key metrics such as data bandwidth, LinkSpeed, and Link Width, showcasing round-trip PCIe communication.
- Managed and provisioned cloud infrastructure across AWS, Azure, and GCP, including VPC networking, EC2 instance configuration, and S3 storage, ensuring high availability and scalability of deployed applications.

SWE Intern, LightSpeed Photonics *April 2021 - August 2021*

- Automated Yocto OS image builds using shell scripting, reducing built time by 75%.
- Built an E2E MERN application to demonstrate video analytics on custom EdgeAI Acceleration Hardware.

Projects

WayNotify - A notification daemon for Wayland *[C++, CMake, Wayland, SDL2]*

- Developed a background service to listen for desktop notifications via D-Bus and display them on-screen using SDL2 and wlroots, achieving <10 ms latency per notification and full compatibility as a drop-in replacement for `dunst` on Wayland.

CPU Scheduling & File System Management Simulation *[Operating Systems, C++, Linux, Dear ImGui]*

- Built an interactive app using Dear ImGui library in C++ that invokes system calls to read/write files, monitor disk usage, modify files & set permissions and simulated job execution process of CPU scheduling algorithms(FCFS, SJF, SRTF, RR).

RISC-V Cache Architecture & Branch Predictor Implementation *[C, GCC, Assembly]*

- Constructed an out-of-order execution pipeline which supports 1-, 2-, and 4-way set associative caches.
- Implemented 1-level and 2-level dynamic branch predictors (2-bit) using BHT and BHSR tables.
- Compared performance metrics by testing set associativity and branch predictors to identify the best configuration.

NY IntelliNews – Real-time AI-Powered Video News Reader Tailored for New Yorkers *[Llama 4, Tavus, NextJs, NodeJs]*

- At Meta's Llama 4 Hackathon, Built NY IntelliNews using Next.js and Node.js, integrating LLaMA 4 for real-time news summarization and Tavus personas for personalized video delivery with <50ms latency.
- Built an interactive UI for topic selection and optimized backend to generate and render summarized news as dynamic videos.

More Projects

Skills

Programming Languages: C++ (proficient), Shell Script, PHP, JavaScript (prior experience)

Frameworks & Technologies: OpenCL, REST APIs, Electron, Node.js, React.js, SQL, NoSQL

Software Tools & Cloud: Git, Docker, Kubernetes, Jenkins, Slurm Workload Manager, Azure, AWS, GCP

Embedded & Hardware Integration: Linux, Arduino, ESP32, Yocto Project, RTOS, SPI, I2C, UART, Device drivers

Certifications & Achievements

Placed in the **top 5%** among 4,783 teams in the **National Cyber League 2025 Team Challenge**.

Completed **Embedded Systems Bootcamp** on Udemy, covering RTOS, IoT, AI, Computer Vision, and FPGA fundamentals.

Earned certificate in **Docker Mastery with Kubernetes and Swarm** from a Docker Captain on Udemy.

Completed **The Complete Web Development Bootcamp** by Angela Yu on Udemy, covering JavaScript, Node.js, ReactJs, etc.